

Grade 6
Unit 11
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

1. Evaluate the expression $x^3 \div r \cdot (k^2 - 6)$ for $k = 4$, $r = 6$, and $x = -3$?

$$\begin{aligned} &= (-3)^3 \div 6 \cdot (4^2 - 6) &= -27 \div 6 \cdot 10 \\ &= (-3)^3 \div 6 \cdot (16 - 6) &= -4.5 \cdot 10 \\ &= (-3)^3 \div 6 \cdot (10) &= -45 \end{aligned}$$

2. Evaluate $r \cdot w \div x \div m$ when $m = 12$, $r = 0$, $w = 4$, and $x = 2$.

$$\begin{aligned} &= 0 \cdot 4 \div 4 \div 2 \div 12 &= 0 \div 12 \\ &= 0 \div 2 \div 12 &= 0 \end{aligned}$$

3. Evaluate $(r \cdot w) \div (x \div m)$ for $m = 12$, $r = 0$, $w = 4$, and $x = 2$.

$$\begin{aligned} &(0 \cdot 4) \div (2 \div 12) &= 0 \div \left(\frac{1}{6}\right) \\ &= 0 \div \left(\frac{2}{12}\right) &= 0 \end{aligned}$$

4. Evaluate $d \div k \cdot y \div b$ if $d = 10$, $k = 4$, $y = 2$, and $b = 6$.

$$\begin{aligned} &= 10 \div 4 \cdot 2 \div 6 &= 5 \div 6 \\ &= 2.5 \cdot 2 \div 6 &= \frac{5}{6} \end{aligned}$$

5. Evaluate $d \cdot k \div y \div b$ when $d = 10$, $k = 4$, $y = 2$, and $b = 6$.

$$\begin{aligned} &= 10 \cdot 4 \div 2 \div 6 \\ &= 40 \div 2 \div 6 \\ &= 20 \div 6 \\ &= \frac{20}{6} = \frac{10}{3} = 3\frac{1}{3} \end{aligned}$$

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 6$ and $y = 2$.

$$= 7 - (2 + 12)^2 + 6(6) \quad = -189 + 36$$

$$= 7 - (14)^2 + 36 \quad = -153$$

$$= 7 - 196 + 36$$

7. Find the value of $r^3 \div x \cdot c$ if $r = 6$, $x = 2$ and $c = 0$.

$$= 6^3 \div 2 \cdot 0$$

$$= 216 \div 2 \cdot 0$$

$$= 108 \cdot 0$$

$$= 0$$

8. Evaluate the expression $3 \cdot (u^2 - 4) + \frac{k}{2}$ for $u = 4$ and $k = 4$.

$$= 3 \cdot (4^2 - 4) + \frac{4}{2} \quad = 36 + 2$$

$$= 3 \cdot (16 - 4) + 2 \quad = 38$$

$$= 3 \cdot 12 + 2$$

9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 1$ and $h = 2$?

$$= 5(1) \cdot 2^3 + (2 - 1)^2$$

$$= 5 \cdot 8 + (1)^2$$

$$= 40 + 1$$

$$= 41$$

10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 2$ and $y = 5$?

$$= \left(\frac{2^2-6}{4}\right) \cdot 7 + 5^2 \quad = -\frac{1}{2} \cdot \frac{7}{1} + 25$$

$$= \left(\frac{4-6}{4}\right) \cdot 7 + 25 \quad = -3.5 + 25$$

$$= \left(\frac{-2}{4}\right) \cdot 7 + 25 \quad = 21.5$$